



NASA AI/ML STIG • 28 SEPTEMBER 2026

Agentic Coding and Research Tools

AI Agents That Choose Instead of Writing

JEV in web browsers, and uses in astronomy

Serat Saad



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The screenshot displays an arXiv search results page for the query "order: -announced_date_first; size: 50; include_cross_list: True; terms: AND author=Serat Saad". The page shows 11 results, with the first two highlighted. The first result is "More than half of recent astronomy papers are written with language-model assistance" by Serat M. Saad and Juan-Sen Ting, announced on September 9, 2026. The second result is "Spectroscopic Binary Detection as Agent-Callable Tools: Detecting 40,000+ Main-Sequence Binary Candidates from SDSS DR19 APOGEE Spectra" by the same authors, announced on September 26, 2026. The sidebar on the right shows a sequence of actions: "DONE", "Find papers whose author is Serat Saad. Stop as soon as the list of search results is showing; do not open any paper.", "Decision time: 668 ms", "Operation: DONE", "DONE 100%", and a list of indexed elements including "Learn more", "Dismiss announcement", "archive home", "Search", "Submit", "Donate", and "Log in".

A run on this laptop: arXiv search for papers by Serat Saad (3 actions, 4.2 s).

Plan for the talk

- 1 How a language model works** writing one word at a time
 - 2 System 1 agents** choosing from an action space
 - 3 JEV** a System 1 agent for web browsers, online
 - 4 Use cases** astronomy websites, galaxies and spectra
-  **Live** the website, and how this was built

1

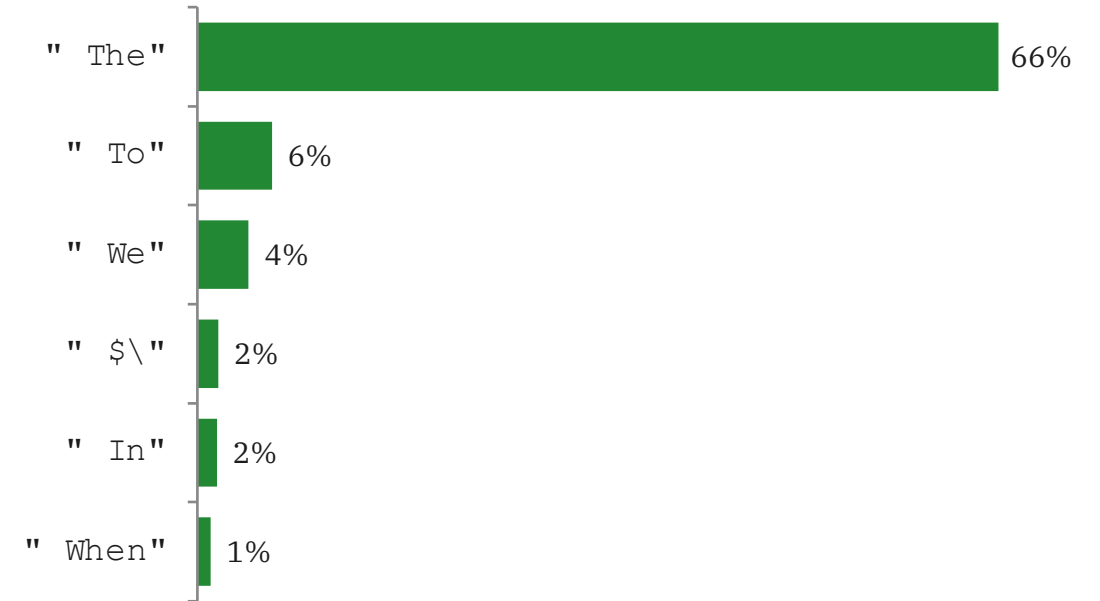
How a language model works

The model writes one word at a time, and each word needs a full pass through the model.

The model predicts the next word

The prompt

You are controlling a web browser on the arXiv advanced search page. The goal is to find papers whose author is Serat Saad. The page offers these controls: [1] link Learn more ... [4] combobox Field to search ... Which control should be used next, and why?
Answer:



Next-token probability, measured

- **One forward pass** reads the prompt and gives a probability for every token (about 150,000).
- **A token** is a word or part of a word.

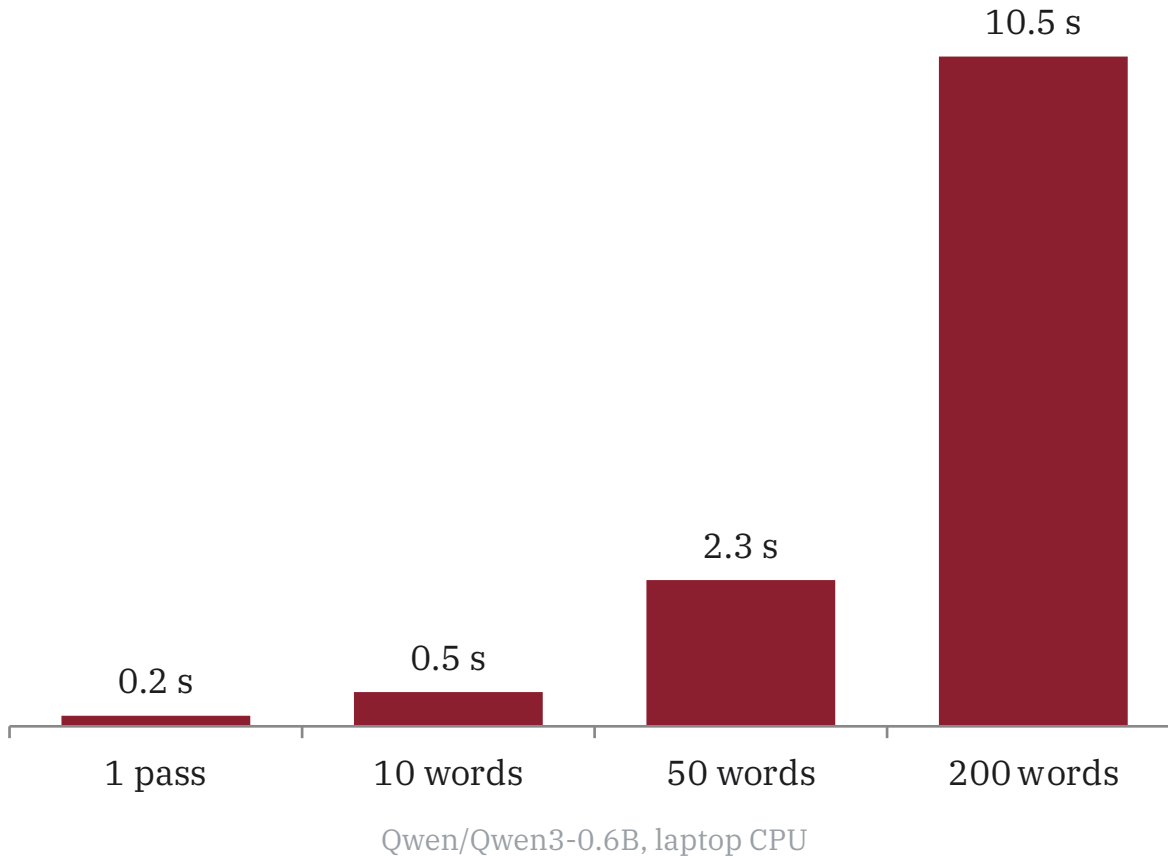
Writing text repeats this step for every word



```
prompt           → model → "The"  
prompt + "The"   → model → "correct"  
prompt + "The correct" → model → "control" ...
```

- **Each new token needs another pass** through the whole model.
- **The model wrote "[4]" on pass 8.** The other passes only add an explanation.

Run time grows with output length



What makes an agent step slow

- **Output length:** about 52 ms per word here
- **Hidden reasoning:** many tokens before the answer
- **Screenshots:** thousands of image tokens per step
- **Repetition:** all of this again for every click

What a general web agent does at every click



71.6 s

median time per task, Browser Use

178k

tokens per task (median)

The action is often just

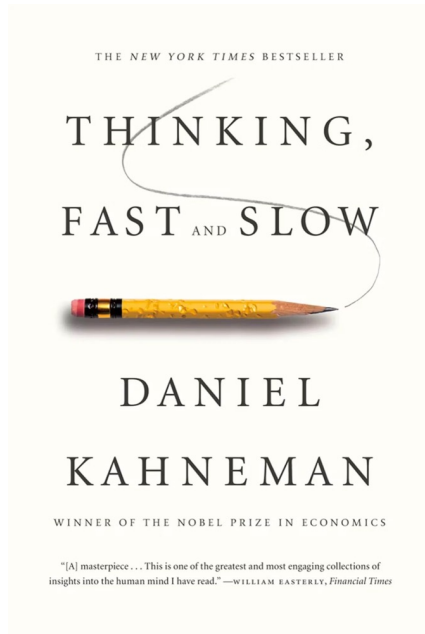
click Search

2

System 1 agents

An agent that chooses its next action from a list, instead of writing it.

Kahneman: thinking, fast and slow



System 1

- fast and automatic
- finds the Search button
- picks from what is in front of it

System 2

- slow and deliberate
- decides how to analyse a data set
- reasoning models

A System 1 agent makes most of its decisions by choosing from a list.

An action space is what the agent can do next

FIXED

Galaxy classifier

3

spiral, elliptical, merger

FIXED

Atari agent

18

joystick moves

FIXED

Writing text

~150,000

tokens, at every word

DYNAMIC

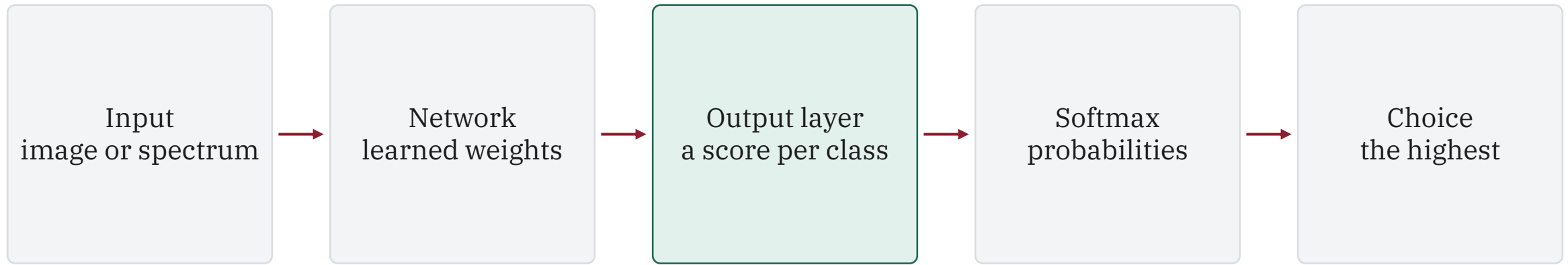
arXiv web agent

7 × 28

operations × controls on
this page

Fixed action spaces are set in advance; dynamic ones change at every step.

How a classifier chooses from a fixed action space



The output layer has one unit per class, so the action space is fixed at training.

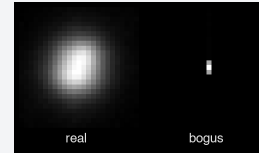
Choosing from a list is classification



Star or galaxy?

Point sources or extended sources.

Image: SDSS



Real or bogus?

Filter detections before a human looks.

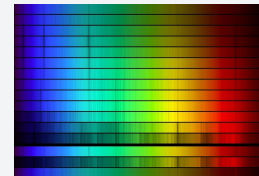
Image: ESO archive, WFI



Which morphology?

Spiral, elliptical or merger.

Image: SDSS



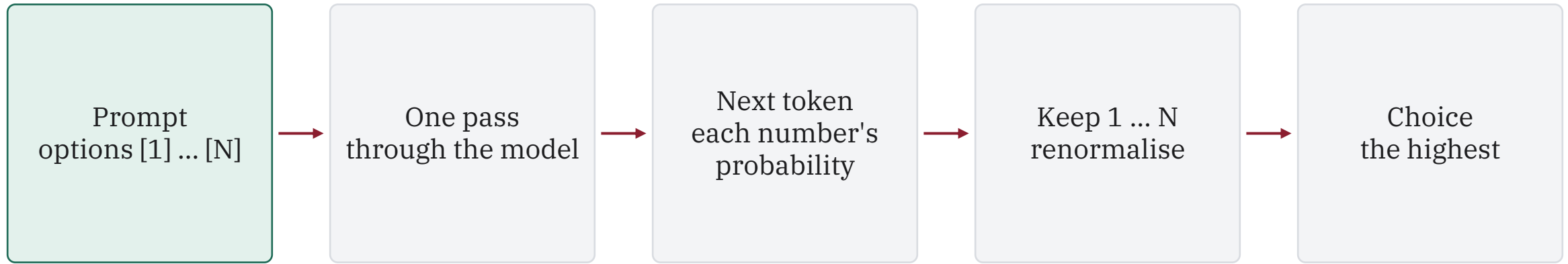
Which spectral type?

One class from a spectrum.

Image: NOIRLab/NSF/AURA

A classifier picks one label from a fixed list, in one pass.

A language model can score any action space



The options are written into the prompt, so the action space can change at every step.

No new output layer and no training.

Time to write and time to choose

$$t_{\text{write}} \approx t_{\text{prompt}} + n t_{\text{token}}$$

Writing: one pass for every token, so the time grows with the length n of the answer.

$$t_{\text{choose}} \approx t_{\text{prompt}}$$

Choosing: one pass over the prompt, however many options there are.

On this laptop: 2.3 s to write 50 tokens, 0.17 s to choose.

The same model, writing and choosing

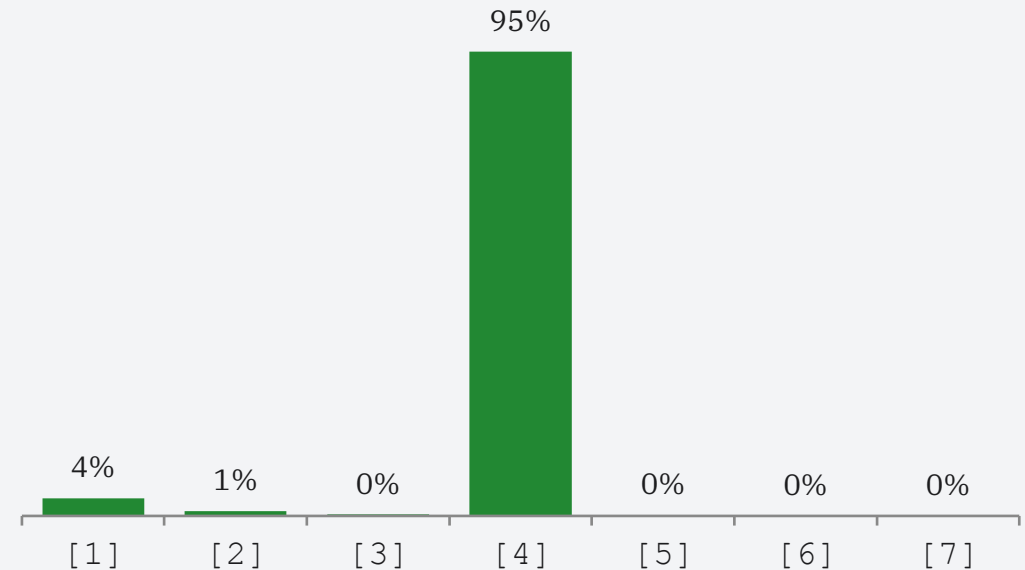
WRITE

2.3 s for 50 words

"The correct control to use next is [4] combobox Field to search. This allows the user to select the author's name, which is what we need to search for..."

CHOOSE

0.17 s one pass



Both give the same answer, [4]. Reading the probability of each option's number is about 13× faster than writing a sentence, and it also gives a probability for each option.

Related ideas



Multiple-choice benchmarks

Each lettered answer gets a probability.



Pointer networks

Point at one input item.



Game agents

A fixed set of legal moves.



Structured outputs

Only valid values can be produced.



Search ranking

The best-scored candidate is returned.



What is new

A new action space from each page.

³

JEV: a System 1 agent for web browsers

A new action space for every page.

Why is it called Jev?



William Stanley Jevons

The Coal Question, 1865

Watt's engines used less coal per unit of work, but Britain's total coal use went up.

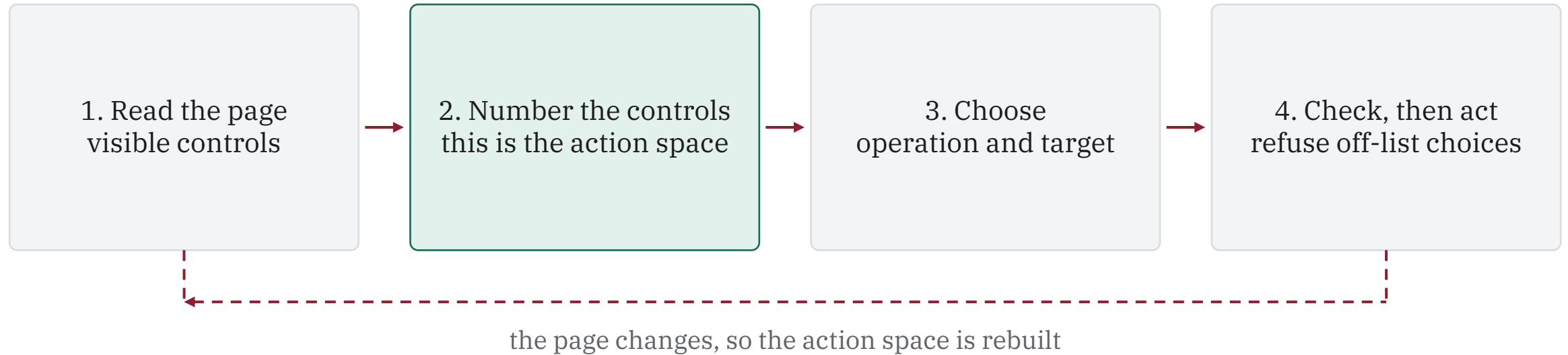


The Jevons paradox

Cheaper use of a resource often raises its total use.

TypeSafe's founder chose the name for this reason.

The loop: a new action space after every action



The loop stops when the model chooses DONE.

JEV's action space at one step

Operations (7)

CLICK
TYPE_TEXT
SELECT
SCROLL
WAIT
DONE
BLOCKED

×

Targets

the numbered controls on this page

28

on arXiv's advanced search

The operations are fixed; the targets change with every page. Only CLICK, TYPE_TEXT and SELECT need a target.

JEV compared with other ways of choosing

	Classic classifier	Zero-shot model	JEV	Generative agent
Action space	fixed at training	listed in the prompt	controls on the page	every token
Training	thousands of labels	none	none	none
Cost per decision	one pass	one pass	one request	one pass per word

Step 1: each page becomes a numbered action space

The screenshot shows the arXiv Advanced Search page. Numbered elements (1-28) are placed over various UI components: 1. Learn more link; 2. Search button; 3. Submit button; 4. Donate button; 5. Log in button; 6. Search term(s) label; 7. Search term input field; 8. Title dropdown; 9. Add another term+ button; 10. Search button; 11. Subject label; 12. All classifications will be included by default; 13. Computer Science (cs) checkbox; 14. Physics checkbox; 15. Economics (econ) checkbox; 16. Quantitative Biology (q-bio) checkbox; 17. Electrical Engineering and Systems Science (eess) checkbox; 18. Quantitative Finance (q-fin) checkbox; 19. Mathematics (math) checkbox; 20. Statistics (stat) checkbox; 21. Include cross-listed papers checkbox; 22. Exclude cross-listed papers checkbox; 23. Date label; 24. All dates radio button; 25. Past 12 months radio button; 26. Specific year radio button; 27. Year input field; 28. Simple Search button.

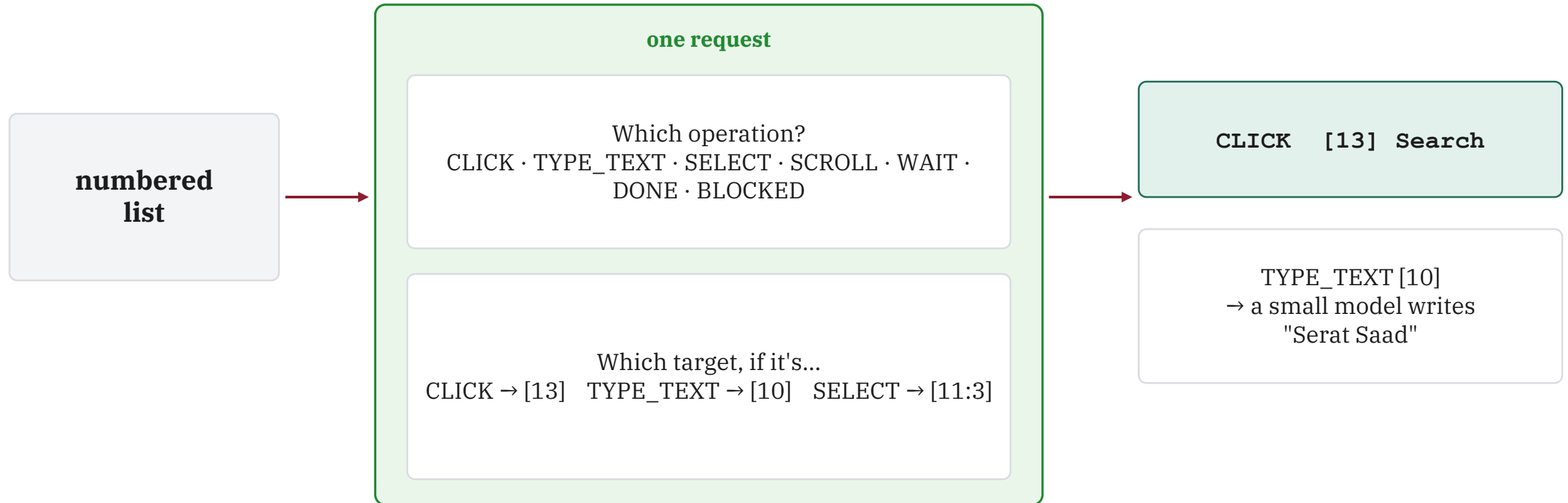
NEXT ACTION		28 elements
Choose an action		
1 Find papers whose author is Serat Saad. Stop as soon as the list of search results is showing; do not open any paper.		
Decision time	Target confidence	
—	—	
Operation		—
Indexed elements		
Unranked		
[1]	Learn more	—
	link · CLICK	
[2]	Dismiss announcement	—
	button · CLICK	
[3]	archive home	—
	link · CLICK	
[4]	Search	—
	link · CLICK	
[5]	Submit	—
	link · CLICK	
[6]	Donate	—
	link · CLICK	
[7]	Log in	—
	link · CLICK	
Operation and target are separate choices in one request. Text is generated only for TYPE_TEXT.		

What the model is sent

```
[10] textbox    Search term · empty
[11] combobox   Field to search · Title
[12] button     Add another term
[13] button     Search
[17] checkbox   Physics
...28 controls in all
```

- only controls that are visible and enabled
- names taken from labels and ARIA, not from pixels
- at most 250 controls and 6,000 characters of visible text
- no screenshot is sent to the model

Step 2: one request asks two questions



- **Only TYPE_TEXT needs written text**, and a small separate model writes it.

Step 3: the agent acts only within the action space



Offered numbers only

Other indexes are refused.



Page unchanged

Otherwise it decides again.



Target clickable

Covered buttons are refused.



No model code

The model never writes code.

An option that is not on the list is refused before anything is clicked.

Where to get each part of JEV



AVAILABLE

The loop and inspector

jev-ultrafast

github.com/browser-use/jev-ultrafast

MIT licence, runs locally.



OPEN NOW

The decision model

TypeSafe API

typesafe.ai

Closed while I made this talk.



AVAILABLE

An open alternative

AnyJev

github.com/nokia-applied-research/AnyJev

Apache-2.0, any open model.

My tests used JEV's loop with gpt-4.1-mini; TypeSafe's model was closed.

AnyJev: the same idea with open models

AnyJev (Nokia) reads option probabilities from any open model and corrects two known biases.

It runs on this laptop.

Order bias

Answers that change when the options are reversed

23%

raw

7.3%

after correction

Reported by AnyJev

Community projects from the first week



Jev plays Doom

TypeSafe's founder



Jev plays chess

Maxim Saplin



A drone simulator

Roman Slack



A painter

jevinci



Jev plays Tetris

Alan Daitch



Smash Bros.

Mau Baron



Wikipedia link race

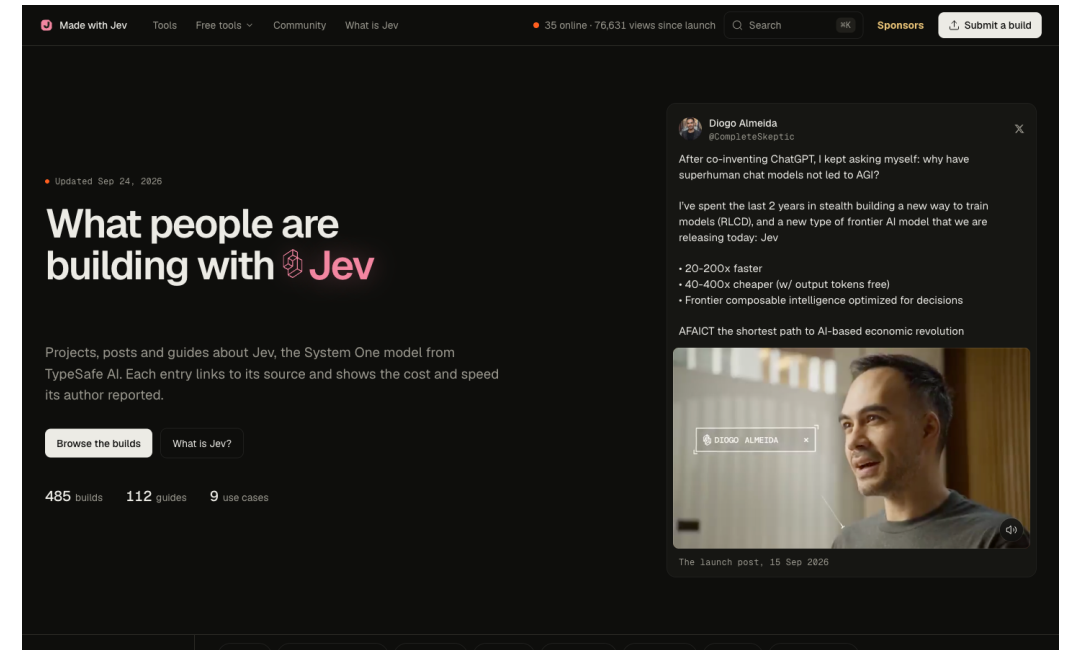
jev-bfs



Email triage

Riley Brown

Reported by the authors (madewithjev.com).



4

Use cases

Astronomy websites first, then choices outside the browser.

Astronomy tasks: what I had to fix in JEV



Gaia: search tab not listed

→ **list skipped elements**



arXiv: needs the Enter key

→ **add PRESS_ENTER**



ESO: links open new tabs

→ **keep the same tab**



SIMBAD: left the answer page

→ **"goal met?" check**



Repeated refused choices

→ **withdraw them**

11

loop changes

Model unchanged

Astronomy tasks: close to Browser Use, 6× faster

Thirteen new tasks, written before any fix. A fixed check on the final page decides success.

JEV's loop with fixes

77%

verified (17/22)

6.3 s

median time

15k

tokens

Browser Use

82%

verified (9/11)

41.1 s

median time

145k

tokens

95% intervals: 57%–90% and 52%–95%

Use cases for a web agent



Archives without an API

old forms built for people



Survey data portals

one query, many targets



Documentation lookup

large manuals



Bibliography checks

ADS entries for references



Literature watch

daily arXiv and ADS



Catalogue cross-checks

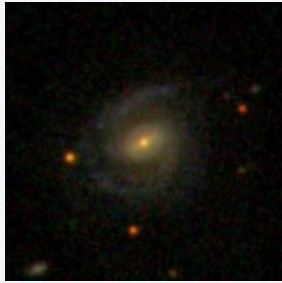
SIMBAD against NED

Use an API when one exists.

System 1 outside the browser: galaxies and spectra

Galaxy morphology

60 SDSS images: spiral, elliptical, merger



65%

choose, 0.41 s a call

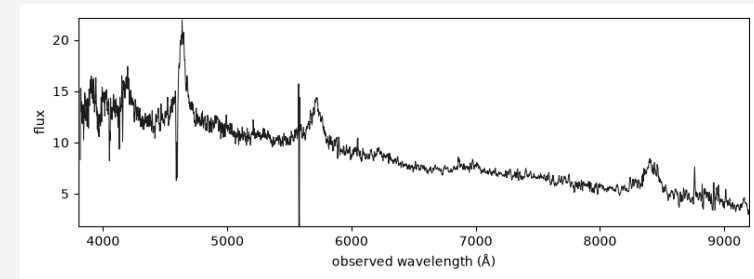
Confident ($p \geq 0.9$): 75% of 24

62%

write, 0.67 s

Spectra

60 SDSS spectra: star, galaxy, quasar



70%

choose, 0.47 s a call

Confident ($p \geq 0.9$): 100% of 15

73%

write, 0.74 s

Choosing is about as accurate as writing, and faster.

Other System 1 decisions in astronomy



Alert streams

real or bogus



arXiv triage

relevance and reader



Data-quality flags

trails, saturation, bad pixels



Catalogue flags

which entries look wrong



Help desks

route tickets to teams



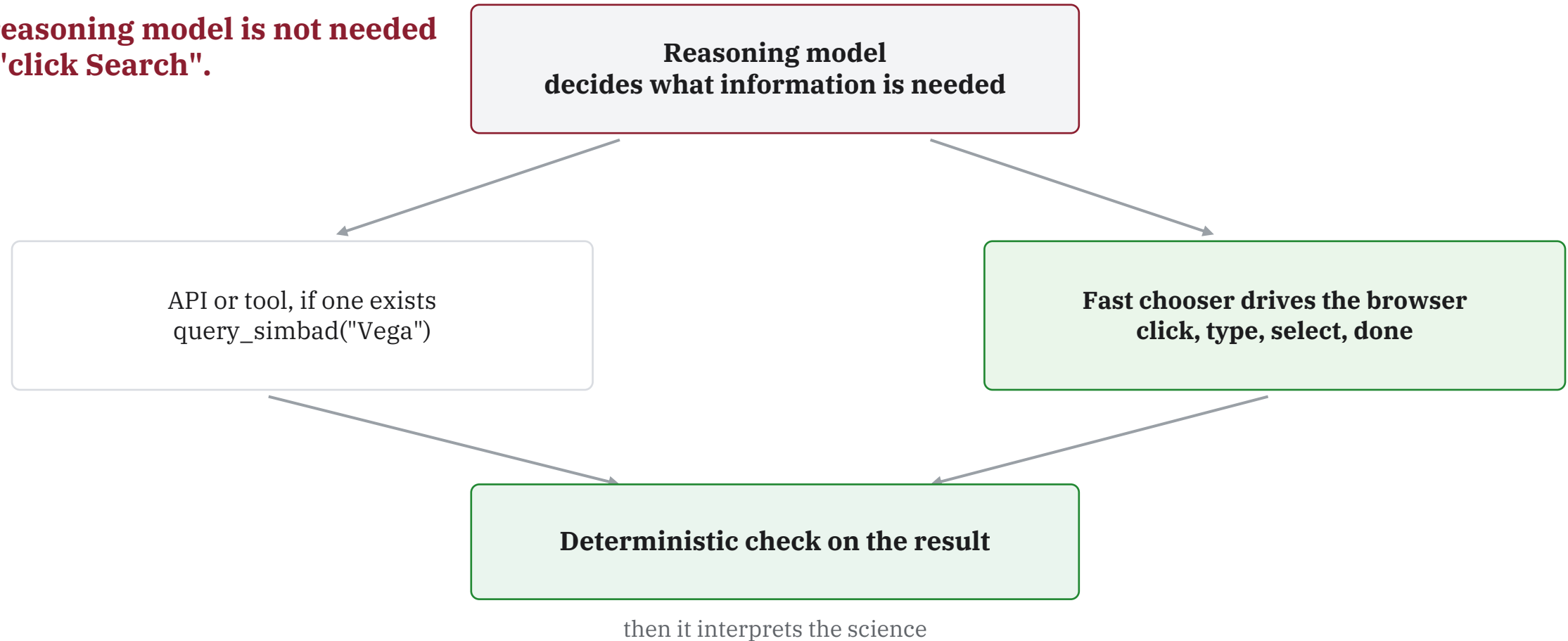
Citizen science

check volunteer labels

Each answer comes from a known set, at high volume.

A possible design: reasoning apart from clicking

A reasoning model is not needed to "click Search".



Back to Jevons



If an archive query takes 6 seconds instead of 41, people will probably run many more queries.

Load

archives built for human traffic

Etiquette

rate limits, caching, bot labels

Provenance

record page, date and version

Checking

checks must scale with volume

What the results show

Shown

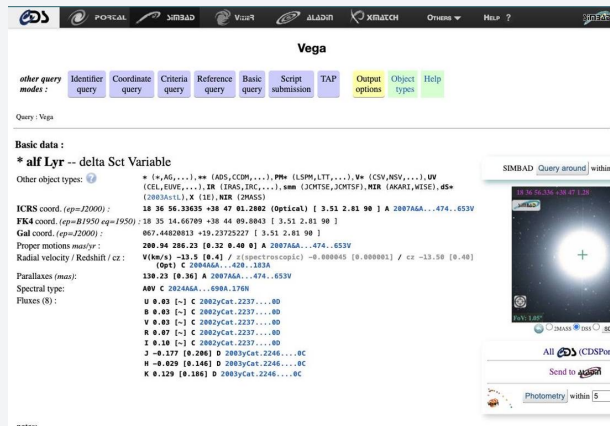
- Choosing from an action space is faster than writing.
- JEV verified 77% of new web tasks, Browser Use 82%.
- On galaxies and spectra, choosing matched writing.

Not shown

- TypeSafe's model, which opened too late.
- That thirteen tasks are enough.
- That a fast agent is a general one.

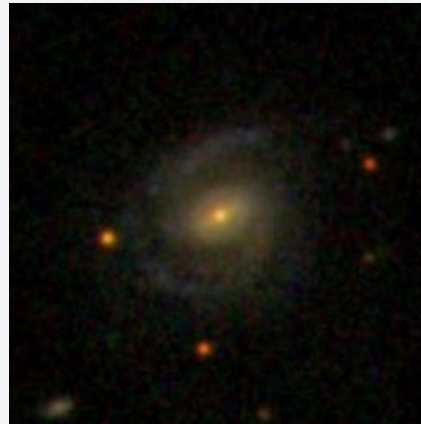
Live: seratsaad.github.io/system1-demos

1 Browser agent



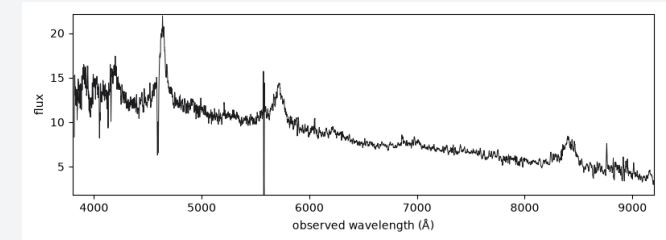
SIMBAD, arXiv and the ESO archive;
recorded runs, replayed at their real
speed

2 Galaxy morphology



spiral, elliptical or merger; choose,
then write

3 Spectra

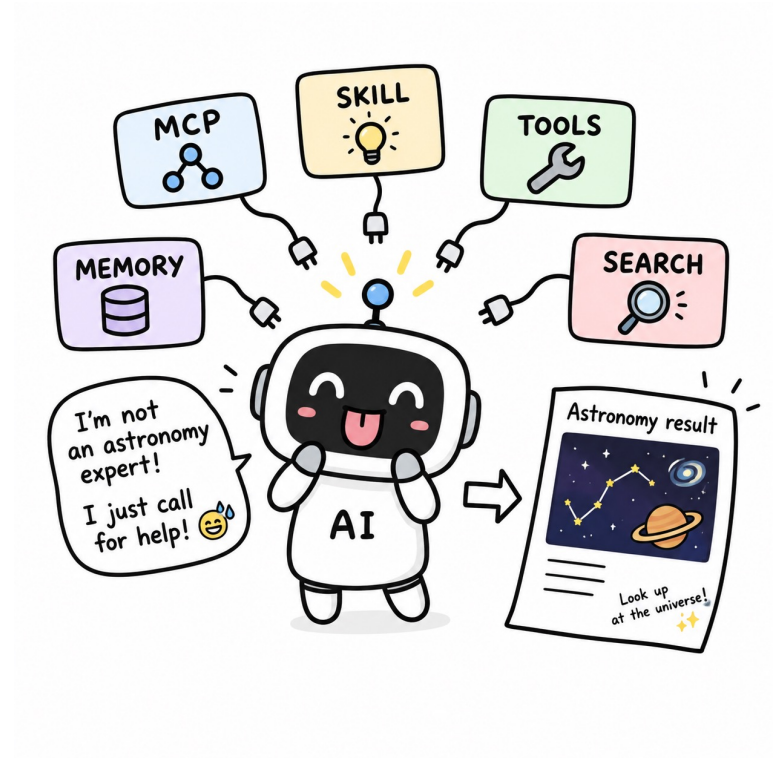


star, galaxy or quasar; choose, then
write

Paste an OpenAI API key at the top of the page. It is not saved.

How this was built

- **The session:** planning, checks and bug fixes
- **Mistakes:** a check on the wrong tab, an expired API key
- **Results:** one JSON file per run, with replay
- **Inspector:** about 150 lines in demo/inspector.py



Many of an agent's decisions are choices from a list. These can be fast if the results are checked, and slower models can be kept for the science.

Serat Saad



All slides are made with AI, but the contents are mine.